Adreslarni tarjima qilish — NAT

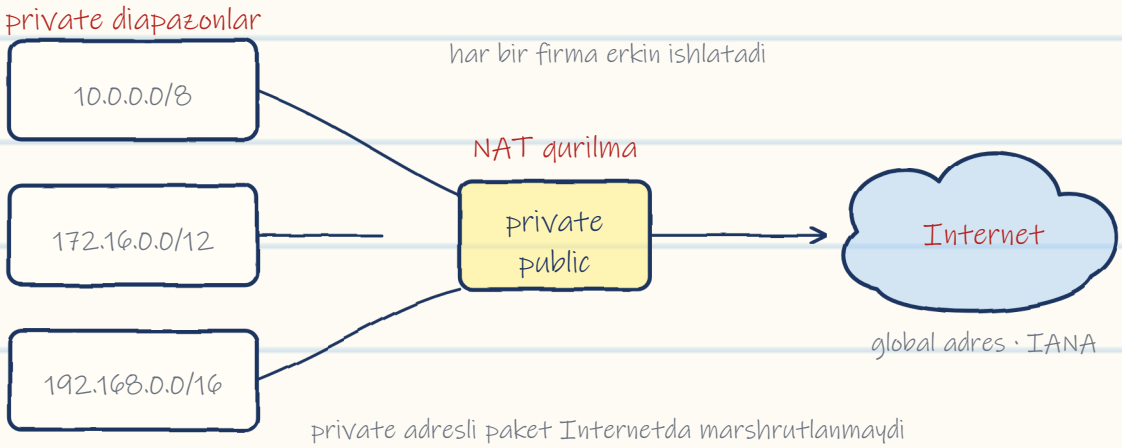
Network Address Translation

Asosiy atamalar

- NAT — private adresni public adresga almashtirish
- Private adres — faqat ichki tarmoqda ishlaydi (RFC 1918)
- NAPT — bitta public adres + portlar — ko'p host uchun
- NAT Server — tashqaridan ichki serverga kirish uchun moslik
- Address pool — tarjima uchun ajratilgan public adreslar

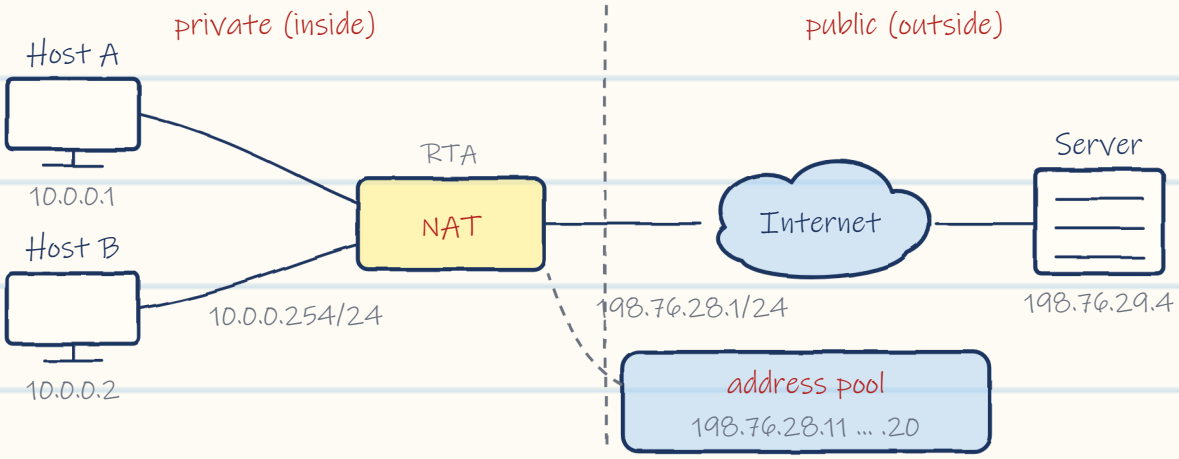
1. Nega kerak

IPv4 tugab bormoqda. Private adres — ichkarida, public adres — Internetda (RFC 1918 · RFC 3022)



1-rasm. Private adres erkin ishlatiladi, chegarada public ga aylanadi

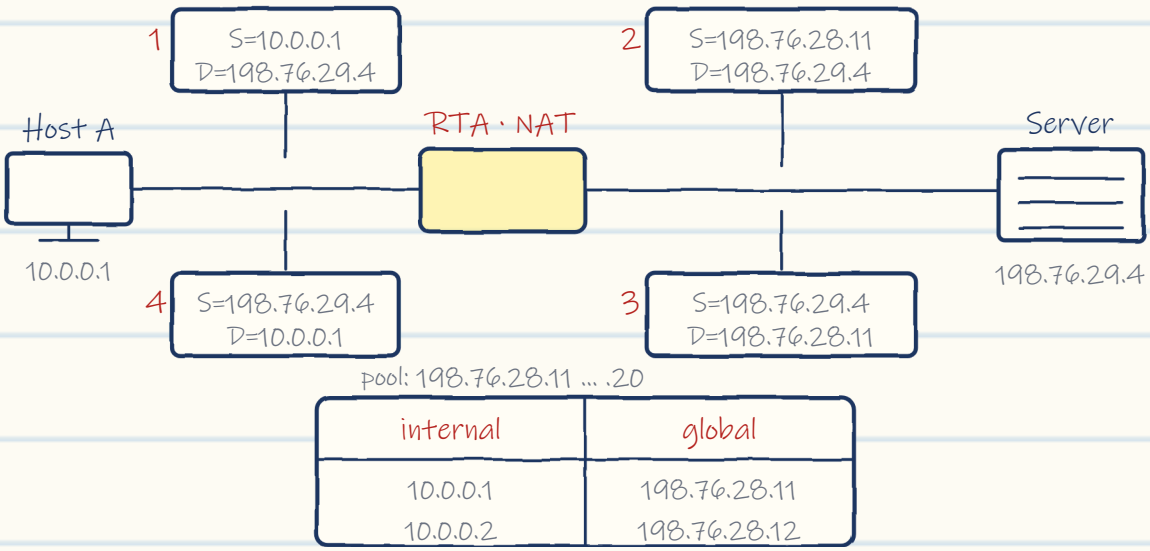
2. Terminlar



2-rasm. NAT qurilma ikki olam orasida turadi

NAT beradi: adres tejaladi + ichki tuzilma tashqaridan ko'rinmaydi.

3. Basic NAT — bir-birga

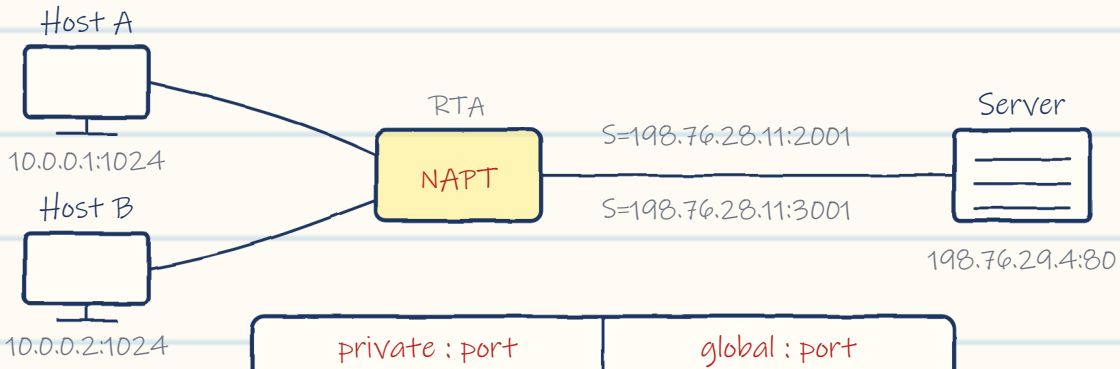


internal	global
10.0.0.1	198.76.28.11
10.0.0.2	198.76.28.12

3-rasm. Faqat manba IP almashadi, NAT jadvaliga yozuv qo'shiladi

- Bitta private adresga — bitta public adres (no-pat)
- Qaytgan paket: maqsad IP jadval bo'yicha ichki adresga qaytariladi

4. NAPT — ko‘p-birga

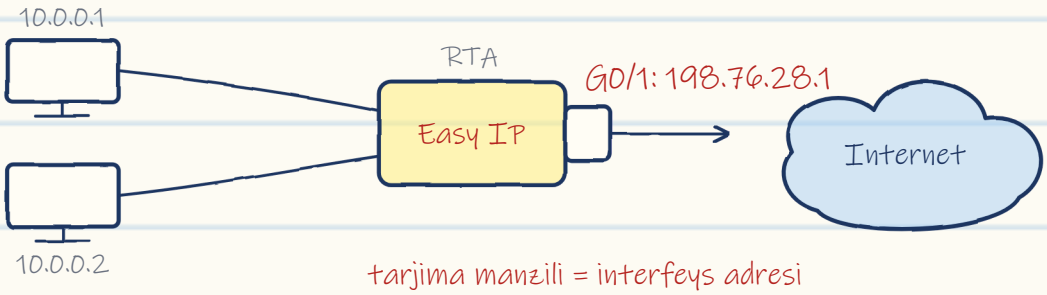


private : port	global : port
10.0.0.1:1024	198.76.28.11:2001
10.0.0.2:1024	198.76.28.11:3001

bitta public adres — ko‘p host, port bo‘yicha ajratiladi

4-rasm. NAPT IP + port juftini tarjima qiladi

5. Easy IP



tarjima manzili = interfeys adresi

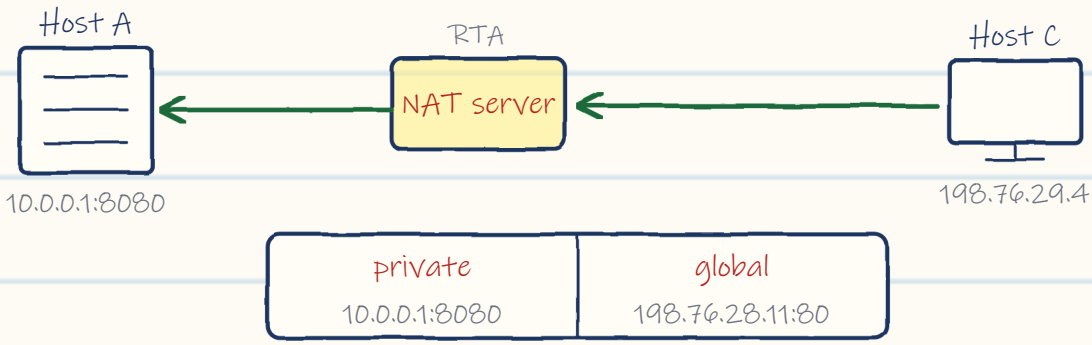
address pool kerak emas · dial-up va dinamik IP uchun

5-rasm. Easy IP — NAPT ning interfeysga bog‘langan ko‘rinishi

6. Internal server

tashqaridan ichkariga — statik moslash

public 198.76.28.11:80 → private 10.0.0.1:8080



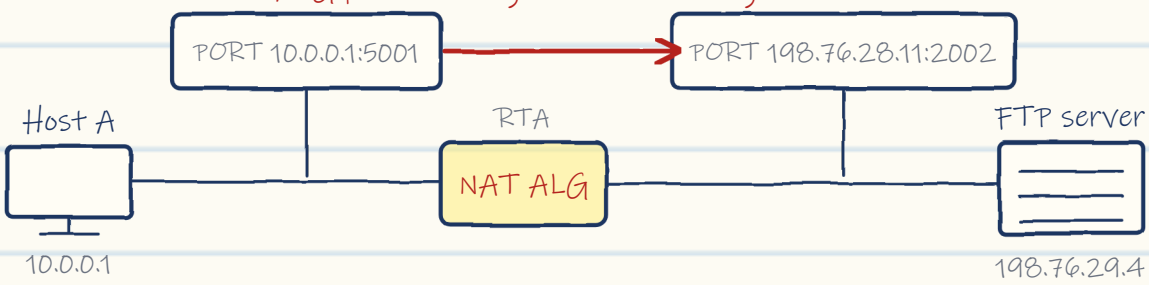
6-rasm. Basic NAT/NAPT da seansni faqat ichki host boshlaydi — bu yerda aksincha

Buyruqlar:

- pool: nat address-group 1 → address 198.76.28.11 198.76.28.20
- basic NAT: nat outbound 2000 address-group 1 no-pat
- NAPT: nat outbound 2000 address-group 1 · Easy IP: nat outbound 2000
- server: nat server protocol tcp global 198.76.28.11 telnet inside 10.0.0.1 telnet

7. NAT ALG

ALG paket ichidagi adresni ham o'zgartiradi



boshqaruv kanali TCP 21 · data kanali TCP 20

FTP · DNS · H.323 · SIP

oddiy NAT faqat IP sarlavhasini ko'radi

7-rasm. ALG bo'lmasa FTP ning data kanali ochilmaydi

- Tekshirish: display nat all · display nat session · reset nat session
- NAT — vaqtinchalik yechim; IPV6 da kerak emas

Xulosa

- private: 10.0.0.0/8 · 172.16.0.0/12 · 192.168.0.0/16
- Basic NAT — bir private : bir public (no-pat)
- NAPT — ko'p private : bir public, port bilan
- Easy IP — interfeys adresi, pool yo'q
- Internal server — tashqaridan ichkariga; ALG — FTP, DNS, H.323, SIP

Tekshirish

1. no-pat nimani bildiradi? — port tarjimasiz, bir-birga
2. Easy IP qaysi adresni oladi? — chiqish interfeysining adresini
3. Tashqi host ichki serverga qanday kiradi? — nat server statik moslash
4. ALG nima uchun kerak? — data ichidagi IP/port uchun